

CSE471 | DEEP LEARNING FOR COMPUTER VISION • WEEK 2, LECTURE 4

Loss Functions & Optimization

MSE, Cross-Entropy Loss, Gradient Descent, SGD, Momentum, and Adam

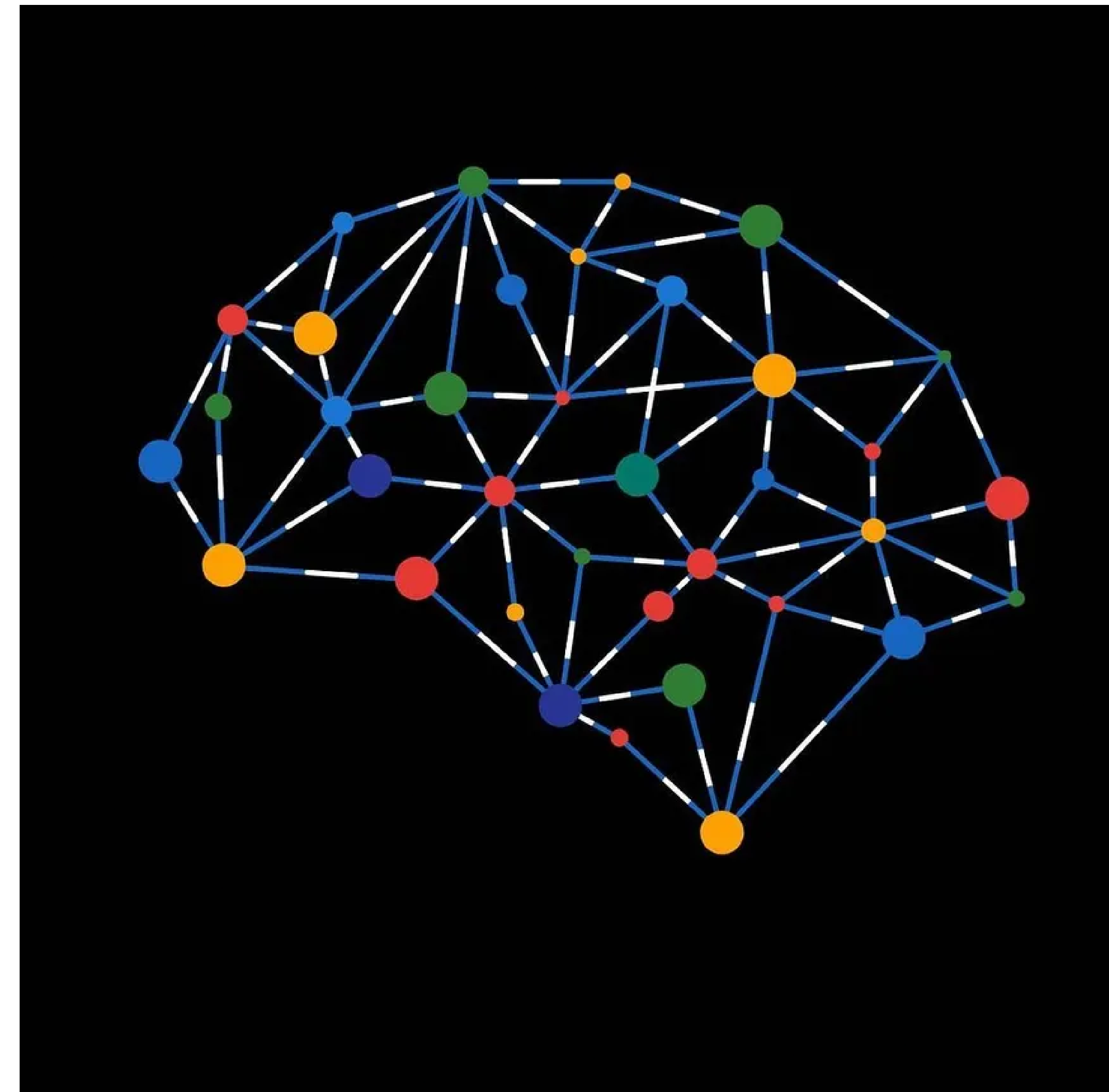
Lovely Professional University

Lecturer - Prof Rishi Raj

Learning Outcomes

By the end of this lecture, you should be able to:

- > Explain why neural networks require loss functions.
- > Differentiate regression and classification losses.
- > Compute Mean Squared Error (MSE) and understand Cross-Entropy.
- > Explain Gradient Descent and compare Batch vs SGD.
- > Understand advanced optimizers (Momentum, Adam).



Lecture Roadmap



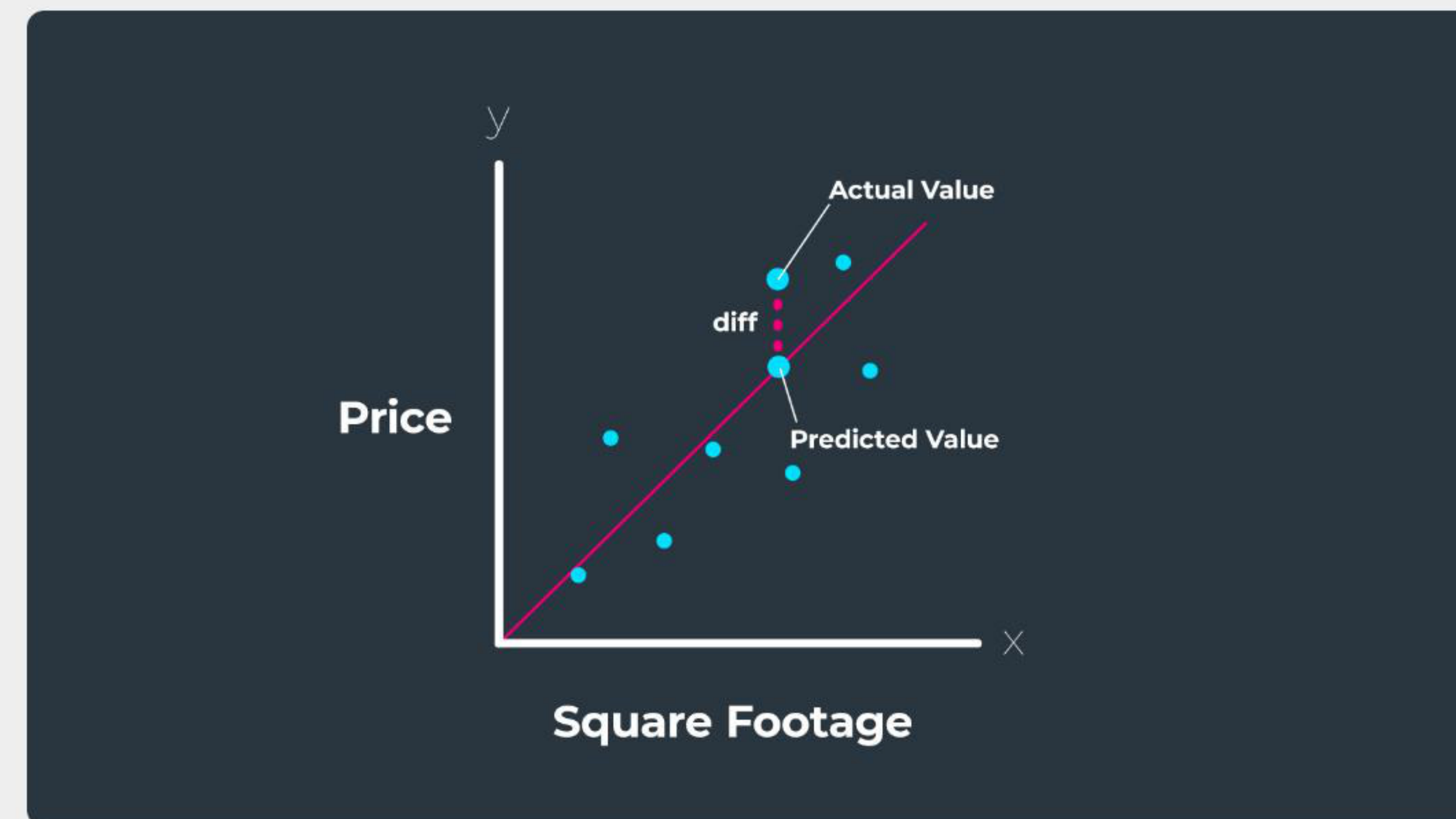
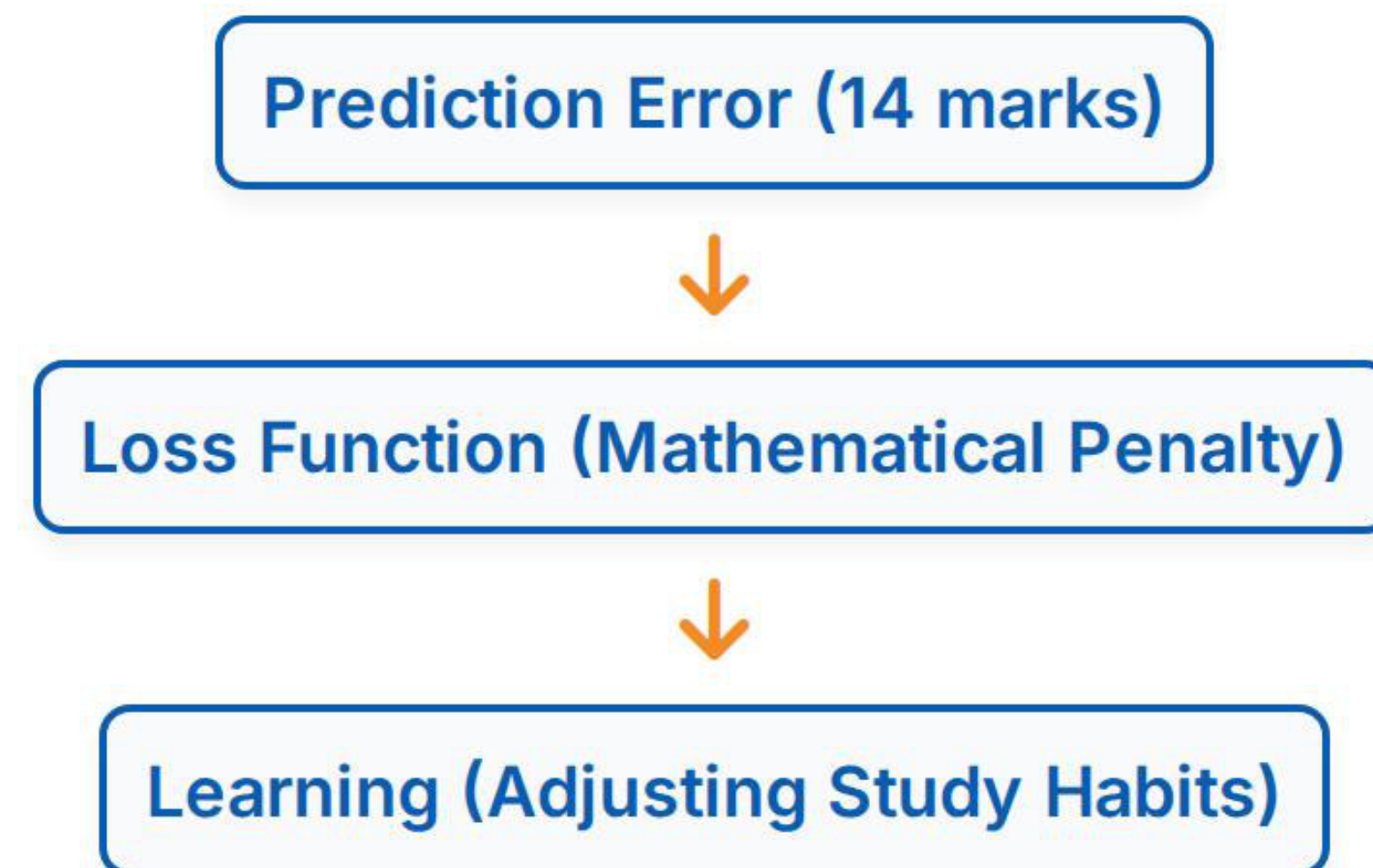
Analogy: Navigation System

Think of this entire roadmap as a GPS. **Prediction** is your current location. The **Loss Function** calculates how far off-route you are. **Optimization** (Gradient Descent) computes the new route, and **Adam/Momentum** are the engines driving you back on track quickly and smoothly.

Quantifying Errors: The Set Induction

Suppose a student predicted they would score 92 marks in an exam but actually scored 78.

How wrong was the prediction?



Why Do We Need Loss Functions?

Without a loss function, a neural network is flying blind. It cannot determine whether its predictions are improving or worsening.



The GPS Analogy

Loss functions act like a GPS recalculating a route. When the network takes a wrong turn (makes a bad prediction), the loss function quantifies the distance off-route so the optimizer knows how to correct the steering.

Regression vs Classification

The choice of loss function depends entirely on the type of problem we are trying to solve.

Category	Regression	Classification
Goal	Predict continuous numerical values.	Predict discrete categories or classes.
Examples	House prices, Temperature, Stock prices, Age.	Cat vs Dog, Face Recognition, Tumor Detection.
Output Format	A single real number (e.g., \$450,000).	Probabilities across classes (e.g., 90% Cat, 10% Dog).
Primary Loss Function	Mean Squared Error (MSE)	Cross-Entropy Loss

Mean Squared Error (MSE)

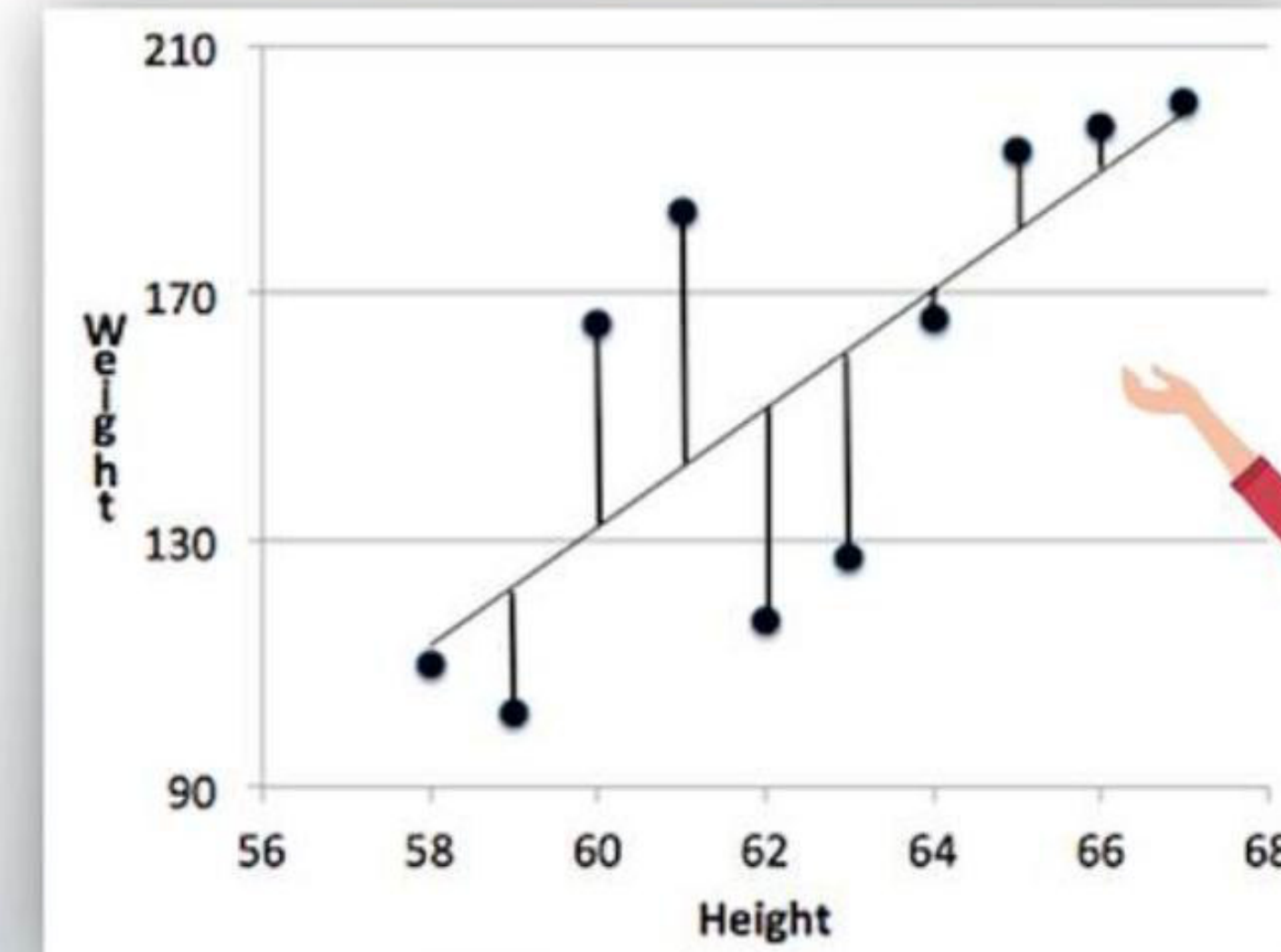
MSE is the standard loss function for regression tasks.

- > **Difference:** Computes distance between actual and predicted values.
- > **Square:** Squares the difference to ensure negative errors don't cancel positive ones, and heavily penalizes large errors.
- > **Average:** Averages across all data points in the batch.

MSE Formula

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

HEIGHT & WEIGHT EXAMPLE



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Numerical Example: Calculating MSE

Let's calculate the MSE for a small batch of 3 predictions.

Data Point (i)	Actual Value (y)	Predicted Value ($\hat{y}$)	Difference (y - $\hat{y}$)	Squared Difference (y - $\hat{y}$) ²
1	2	3	-1	1
2	4	5	-1	1
3	6	4	2	4

$$\text{MSE} = \frac{1+1+4}{3} = \frac{6}{3} = 2.0$$

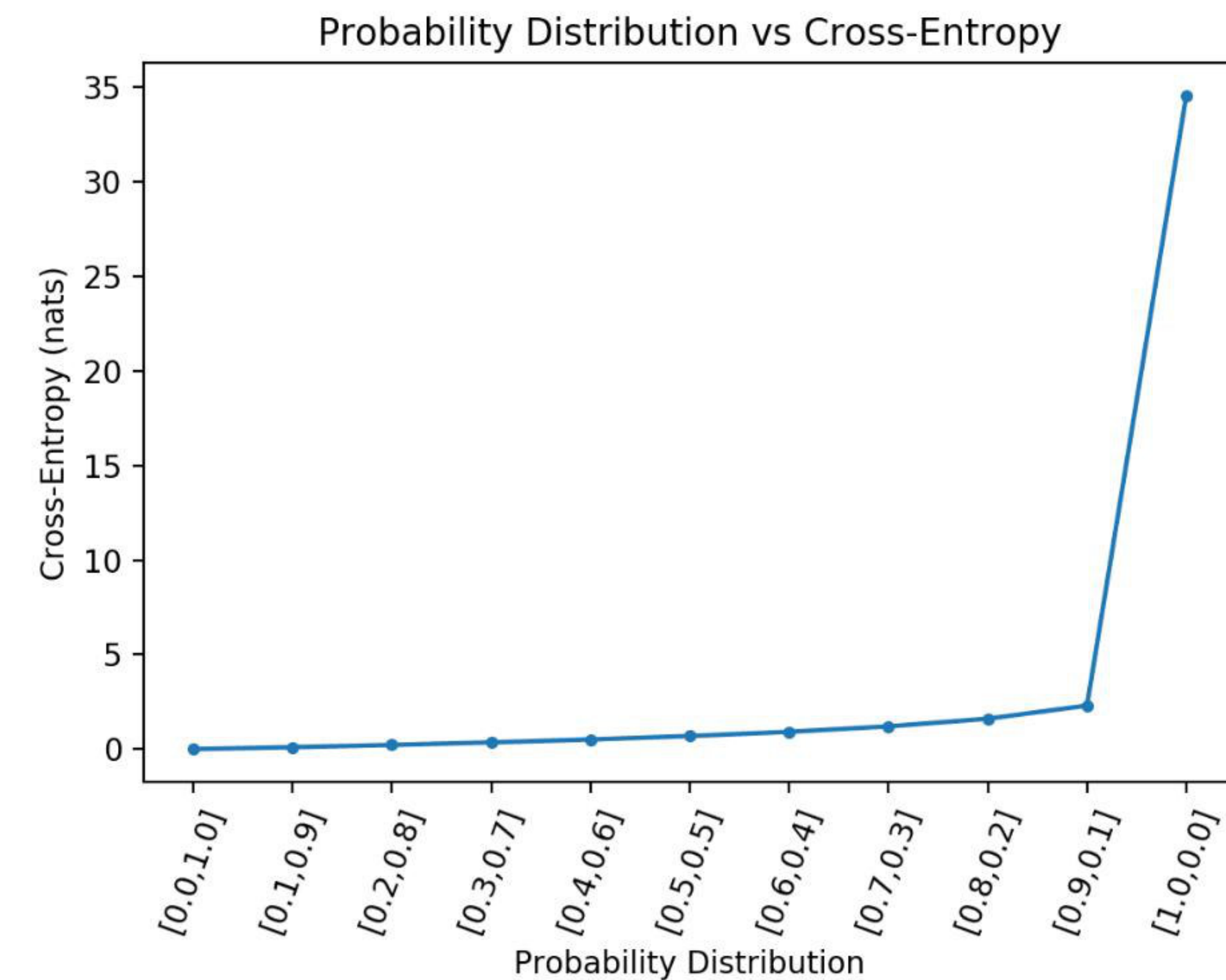
Cross-Entropy Loss (Classification)

Why not use MSE for classification? Because classes (Cat, Dog, Bird) don't have natural numerical distances.

- > Cross-Entropy works with **probabilities** (usually output via Softmax).
- > **Correct & Confident:** Produces a very low loss (near 0).
- > **Incorrect & Confident:** Severely penalizes the network (high loss).

💡 The Information Theory Root

Cross-Entropy comes from Information Theory. It measures the difference between two probability distributions: the true distribution (one-hot labels) and the predicted distribution.



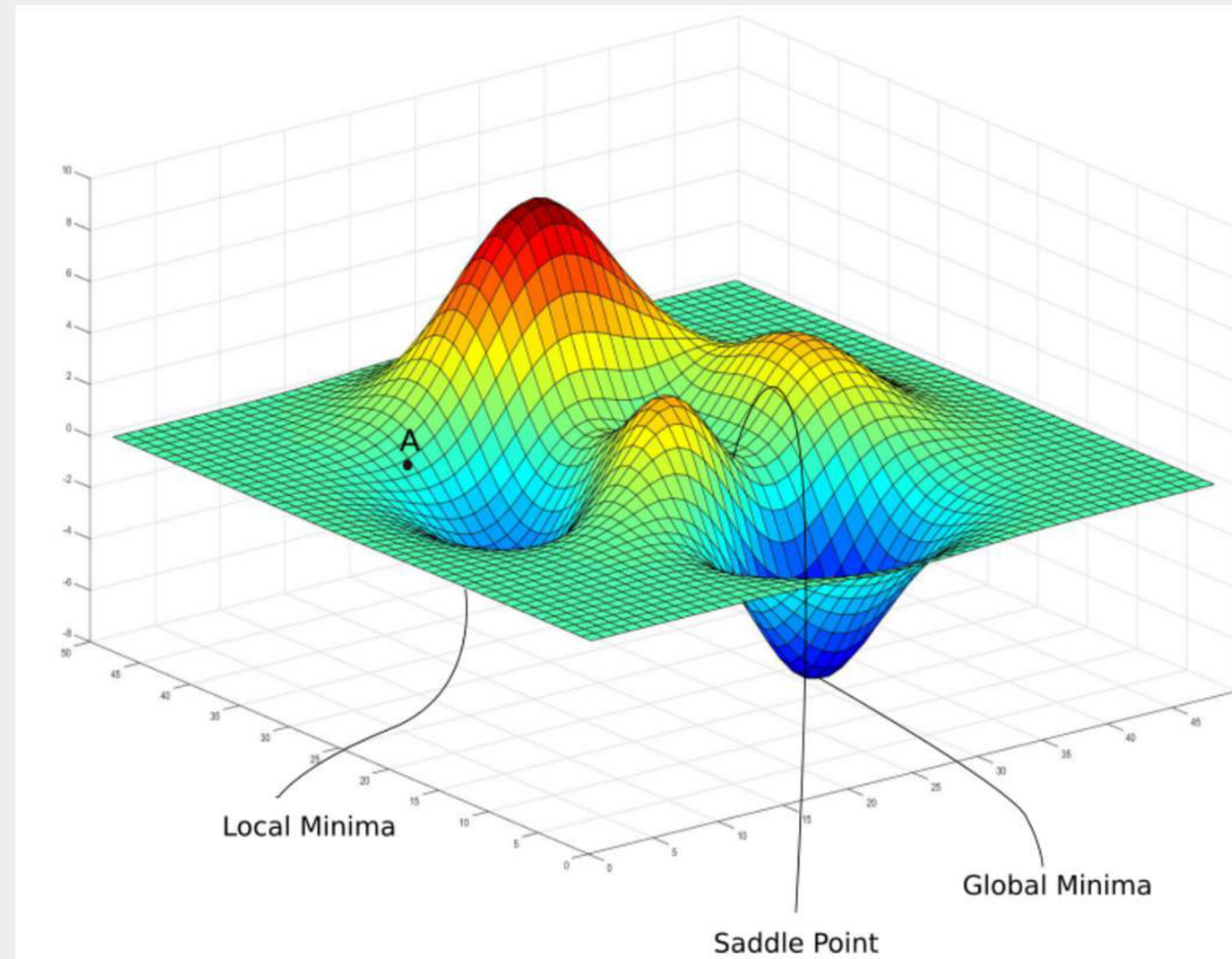
MSE vs Cross-Entropy

Feature	Mean Squared Error (MSE)	Cross-Entropy (CE)
Primary Purpose	Regression (Continuous values)	Classification (Categorical probabilities)
Output Space	Real numbers ($-\infty$ to $+\infty$)	Probabilities (0 to 1)
Mathematical Behavior	Penalizes distance quadratically	Penalizes confidently wrong predictions logarithmically
Gradient Behavior	Can suffer slow learning if far from target	Provides steep gradients when predictions are very wrong, accelerating learning

What is Optimization?

Optimization is the process of finding the model parameters (weights and biases) that minimize the loss function.

- > **The Loss Surface:** Imagine a hilly landscape where height represents the error.
- > **Global Minimum:** The lowest possible valley (lowest error). This is our goal.
- > **Local Minimum:** A shallow valley. The model might get stuck here.
- > **Saddle Point:** A flat plateau that slows down optimization.



Gradient Descent

Analogy: A blindfolded person trying to walk down a mountain to the lowest valley by feeling the slope of the ground under their feet.

$$w_{\text{new}} = w_{\text{old}} - \eta \times \nabla L$$

- > **Gradient (∇L):** The direction of steepest ascent. We subtract it to go down.
- > **Learning Rate (η):** The size of the step we take.
- > **Update:** We repeat this process iteratively until convergence.



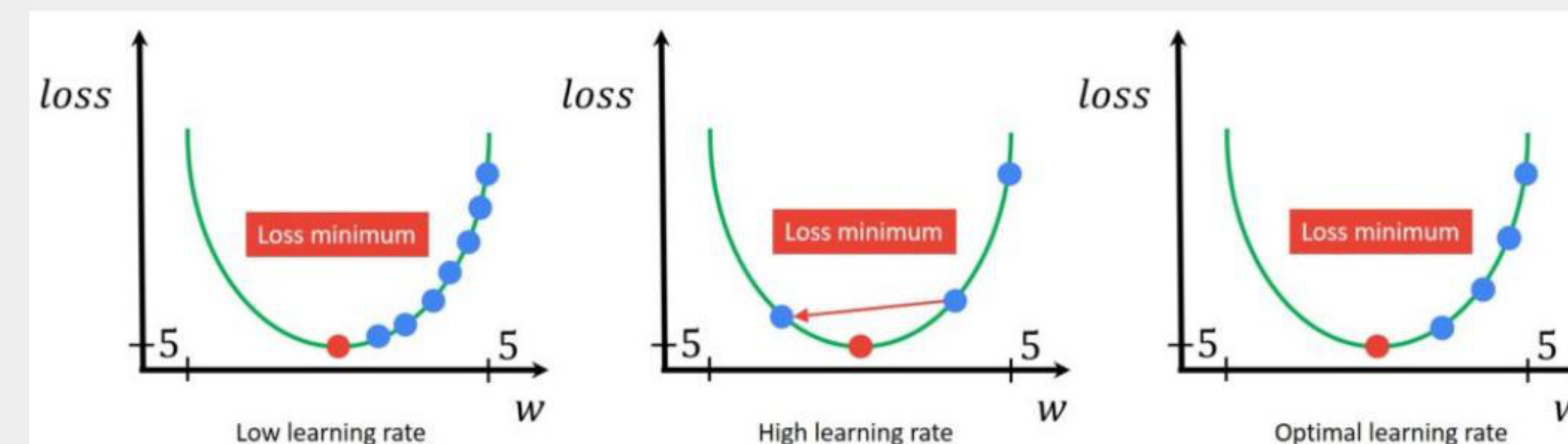
The Role of the Learning Rate (η)

The learning rate is the most critical hyperparameter in optimization.

- > **Too Small:** Network learns reliably but extremely slowly. May get stuck in local minima.
- > **Too Large:** Network overshoots the minimum, oscillating wildly. Loss may diverge (explode).
- > **Optimal:** Smooth and rapid convergence to the global minimum.

Tuning

Finding the right learning rate often involves trial and error, or using "Learning Rate Schedulers" that start high and decay over time.

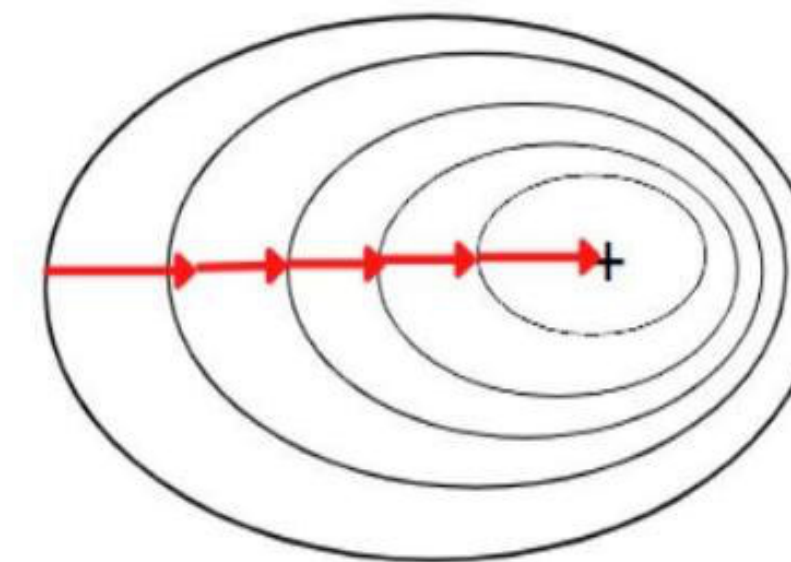


Variants of Gradient Descent

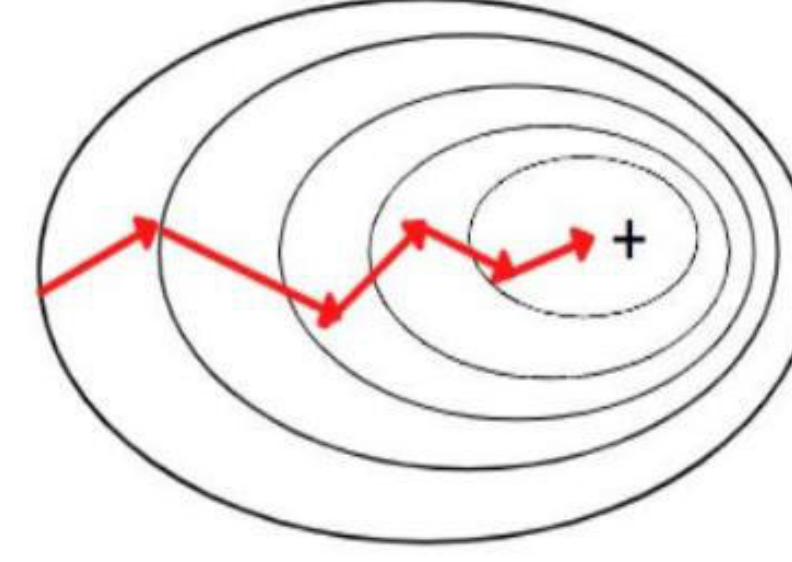
How much data do we look at before taking a step?

- > **Batch GD:** Computes gradient over the *entire* dataset. Slow, memory-heavy, but stable.
- > **Stochastic GD (SGD):** Computes gradient on a *single* sample. Fast, but very noisy trajectory.
- > **Mini-Batch GD:** Computes gradient on a small batch (e.g., 32, 64 images). The industry standard sweet spot.

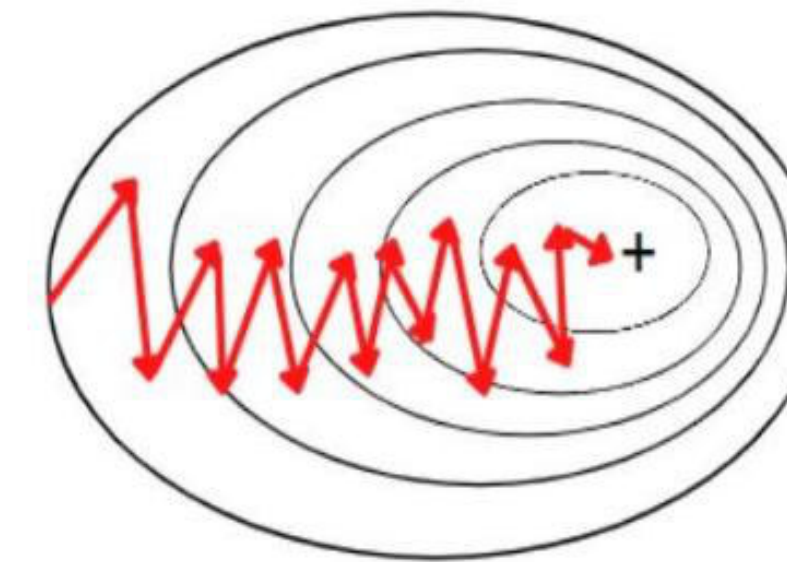
Batch Gradient Descent



Mini-Batch Gradient Descent



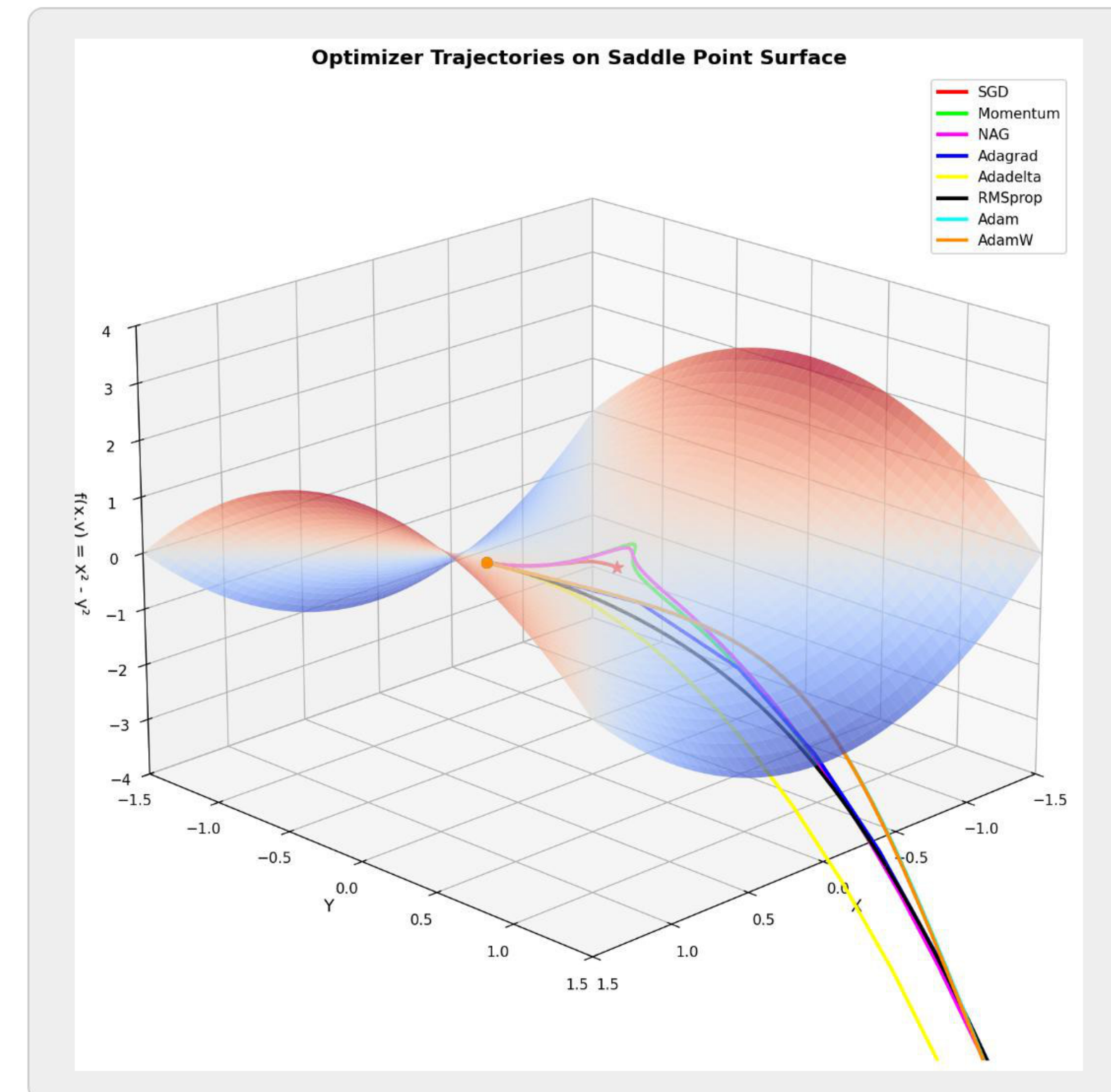
Stochastic Gradient Descent



Momentum Optimizer

Standard SGD can zigzag inefficiently across ravines in the loss landscape.

- **Analogy:** A heavy ball rolling down a hill gathers momentum.
- Momentum remembers the direction of previous gradients.
- It dampens oscillations in irrelevant directions and accelerates in the correct direction.
- Results in much faster and smoother convergence.



Adam Optimizer (Adaptive Moment Estimation)

Adam is widely considered the default optimizer for modern deep learning (PyTorch, TensorFlow).



🏆 Why Adam?

Introduced by Kingma & Ba in 2014, Adam revolutionized deep learning by automatically adjusting the learning rate during training, preventing the need for massive manual hyperparameter tuning. It works exceptionally well out-of-the-box for computer vision tasks.

Optimizer Comparison Matrix

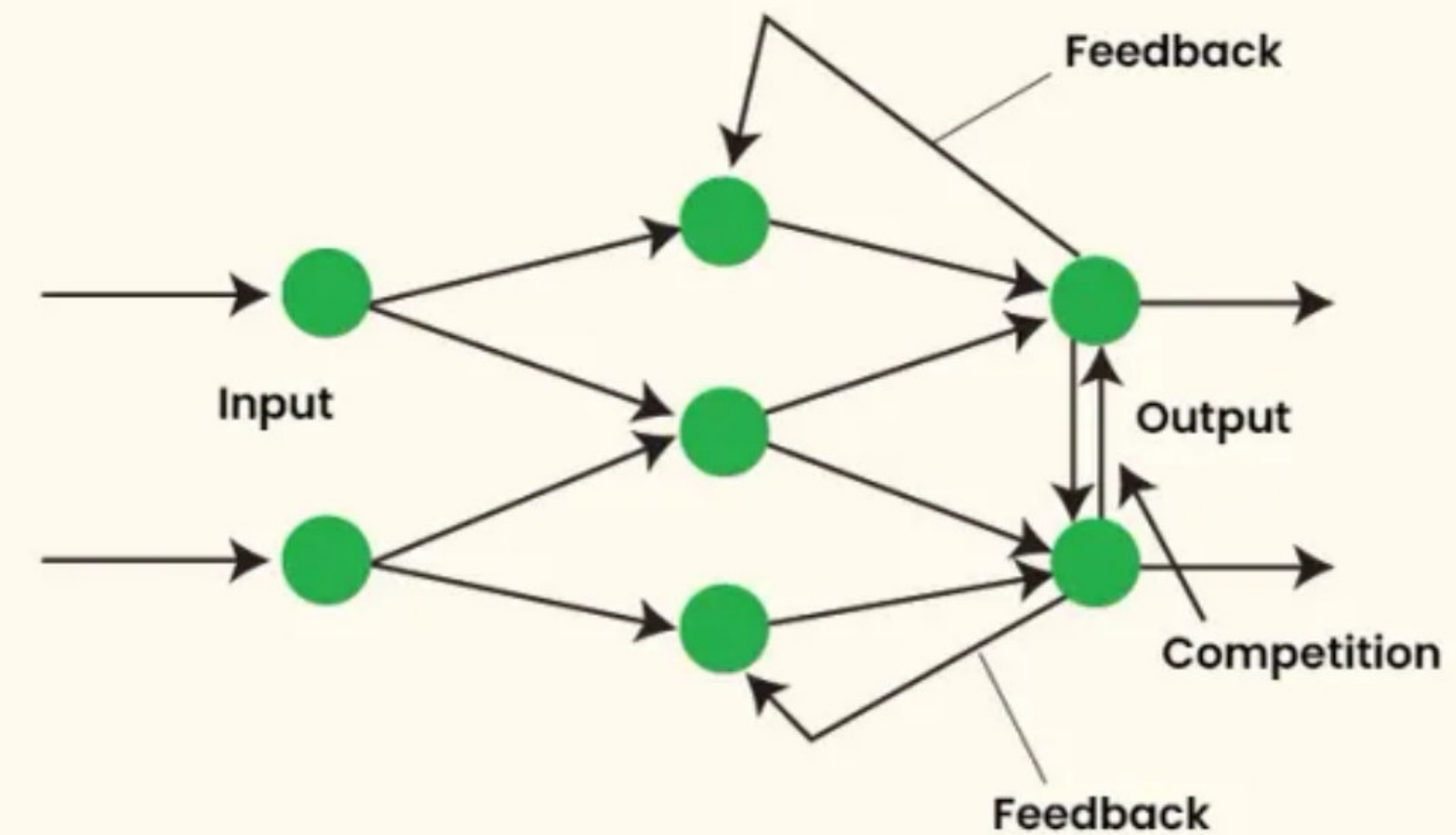
Optimizer	Speed	Stability	Advantage	Disadvantage
Batch GD	Very Slow	High	Exact gradient calculation	Cannot fit in GPU memory
SGD	Fast (per step)	Low (Noisy)	Low memory footprint	Zig-zags, slow to converge globally
Mini-Batch GD	Moderate	Moderate	Good balance of speed and stability	Requires learning rate tuning
Momentum	Fast	High	Escapes local minima easily	Can overshoot the minimum slightly
Adam	Very Fast	Very High	Adaptive learning rates	Computationally slightly heavier

The Full Training Loop

Putting it all together for a Computer Vision model:

- > **Forward Pass:** Pass batch of images through the network to get predictions.
- > **Loss Calculation:** Use Cross-Entropy to measure prediction error against true labels.
- > **Backward Pass (Gradients):** Calculate how much each weight contributed to the loss.
- > **Optimization Step:** Adam uses the gradients to update and improve the weights.

Feedback in Neural Networks



Summary & Key Takeaways

- **Loss Functions:** Essential for quantifying model prediction error (the GPS).
- **MSE vs CE:** MSE is for continuous regression; Cross-Entropy is the standard for categorical classification.
- **Gradient Descent:** The mathematical engine for minimizing loss iteratively.
- **Learning Rate:** Must be carefully tuned to ensure stable, rapid convergence.
- **Optimizers:** Momentum smooths out gradients; Adam combines momentum with adaptive learning rates to become the industry standard.

Think – Pair – Share

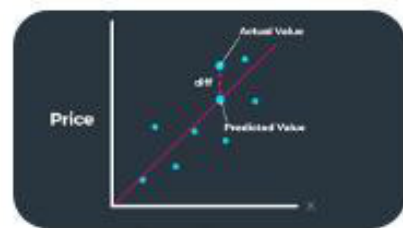
Discussion Question (3 Minutes)

Suppose your neural network's training loss stops decreasing after many epochs (it plateaus). Discuss possible reasons why this is happening.

Homework Assignment:

- Compute the Mean Squared Error for a provided small dataset (Check LMS for values).
- Write a brief explanation of why Cross Entropy is preferred over MSE for image classification.
- Compare SGD, Momentum and Adam using one real-world application for each.

Image Sources



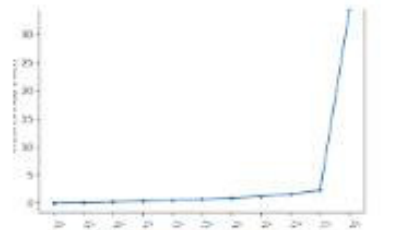
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Source: arize.com



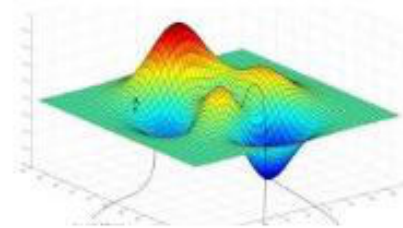
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<https://machinelearningmastery.com/wp-content/uploads/2019/10/Line-Plot-of-Probability-Distribution-vs-Cross-Entropy-for-a-Binary-Classification-Task3.png>

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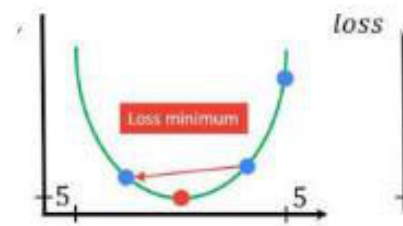
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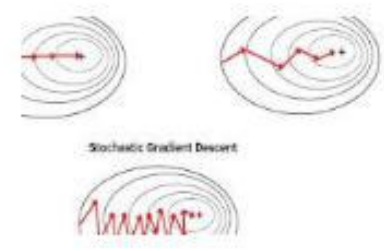
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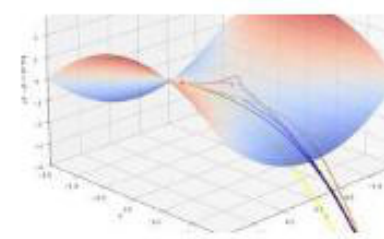
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Image Sources



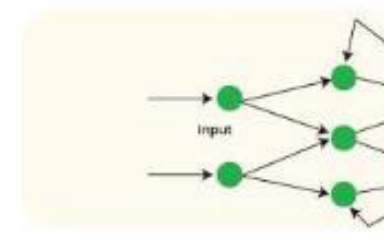
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